American International University-Bangladesh (AIUB)

Department of Computer Science
Faculty of Science & Technology (FST)

Research Methodology

Assignment: Write a research article on the topic of your own choice addressing the following Knowledge Profile (K), Complex Engineering Problem Solving (P), and Complex Engineering Activities (A)

Knowledge Profile (K)
▪ **K3 (Theory based engineering fundamentals):** A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline ▪ **K4 (Forefront specialist knowledge for practice):** Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline ▪ **K5 (Engineering Design):** Knowledge that supports engineering design in a practice area ▪ **K6 (Engineering Practice-Technology):** Knowledge of engineering practice (technology) in the practice areas in the engineering discipline ▪ **K7 (Comprehension of engineering in society):** Comprehension of the role of engineering in society and identified issues in engineering practice in the discipline: ethics and the engineer's professional responsibility to public safety; the impacts of engineering activity; economic, social, cultural, and environmental and sustainability ▪ **K8 (Research Literature):** Engagement with selected knowledge in the research literature of the discipline
Range of Complex Engineering Problem Solving (P)
▪ **P1 (Depth of knowledge required):** Cannot be resolved without in-depth engineering knowledge at the level of one or more of K3, K4, K5, K6 or K8 which allows a fundamentals-based, first principles analytical approach ▪ **P3 (Depth of analysis required):** Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models ▪ **P7 (Interdependence):** Are high level problems including many component parts or sub-problems
Range of Complex Engineering Activities (A)
▪ **A1 (Range of resources):** Involve the use of diverse resources (and for this purpose resources include people, money, equipment, materials, information, and technologies) ▪ **A2 (Level of interaction):** Require resolution of significant problems arising from interactions between wide-ranging or conflicting technical, engineering, or other issues

Submitted By

Semester: Spring_2024-2025				Section:			Group No:			
SL	SN	Student Name	Student ID	Total Marks: 50						
				CO1 (9)	CO2 (9)	CO3 (9)	CO4 (9)	CO5 (9)	Viva (5)	**Total Marks**
A	20	Marufa Islam	22-49431-3							
B	21	Arman Hossain Foej	22-49440-3							
C	22	Joydip Dey Omi	22-49473-3							
D	24	Md. Taqi Yeashir	22-49519-3							

ASSESSMENT & EVALUATION

Evaluation: The assignment will be Evaluated for the following Course Outcomes.

CO1: ***Evaluate*** all relevant resources for designing a computer science and engineering solution and determine the level of novelty of the research.	Total Marks (9)
Problem Analysis and use of State-of-the-Art Resources: Discuss the research problem background with best use of state-of-art literature, resources, and technologies (e.g., related studies & research gap) to produce a significant result that is likely to have a major impact.	[3 Marks]
Critical Reflection and Creativity in Research Objective: Deep insight demonstrated and presented a creative solution to the real-life problem. And Results are critically confronted with various existing literature (e.g., formation of the RQs).	[3 Marks]
Novelty and Contribution of Research: Elaborately discuss and identify the contribution of the research to the development of scientific concepts by recognizing the limitation (e.g., research gaps) of existing research/developments.	[3 Marks]

CO2: ***Analyze*** the collected data to provide valid solutions to the research problem acknowledging the limitations.	Total Marks (9)
Data Analysis: Elaborately discuss the research method, its appropriateness and details on data collection, analysis, and synthesis for proposing valid solution to the research problem (e.g., research methodology)	[3 Marks]
Solution and Validation: Elaborately discuss and validate the solution of the research problem by establishing a direct connection between proposed solutions with the research objective based on the collected research data (e.g., result & analysis).	[3 Marks]
Limitation of the Study: Elaborately discuss the research summary and conclude remarks of the research by acknowledging the limitations of the studies (e.g., abstract and conclusion)	[3 Marks]

CO3: ***Formulate*** and explain ethical constraints and considerations of engineering practice in conducting research to provide computer science and engineering solutions.	Total Marks (9)

Ethics and Societal Impact of Engineering Practice: Understanding engineering's societal role, ethical responsibilities, and the economic, social, cultural, environmental, and sustainability impacts of engineering practice.	[3 Marks]
Approaches to Resolving Ethical Issues in Engineering Research: Provide a detailed discussion and justification of viable approaches for resolving ethical issues, ensuring that ethical principles are clearly articulated and seamlessly integrated within the research context.	[3 Marks]
Compliance with Report Formatting and Submission Guidelines: Submitted in due time, the report is complete and there are no errors in spelling, format, and grammar. Consistently presents a logical and effective organization.	[3 Marks]

CO4: ***Depict*** the need for continuing education and participation in professional societies and meetings.	Total Marks (9)
Motivation of the research: Explore extensively the research topics with evidence rich awareness of the research problems (e.g., facts, citations) indicating intense interest in the area.	[3 Marks]
Comprehension and problem solving: Excellent understanding of material and completely demonstrates effective problem- solving skills integrating alternate and divergent ideas or processes to solve the research problem.	[3 Marks]
Future studies: The scope of future studies is stated and discussed elaborately with details of how this study can be extended in future endeavor.	[3 Marks]

CO5: ***Defend*** the research solutions based on complex engineering activities by delivering an effective presentation to the audience.	Total Marks (9)
Presentation delivery (eye contact and body language): Keeps eye contact with audience all the time, use natural gestures and movements, looks confident.	[3 Marks] **A:** **B:** **C:** **D:**
Enthusiasm/Audience Awareness: Demonstrate strong enthusiasm about the topic, significantly increases audience understanding and knowledge of the topic, convinces an audience to recognize the validity and importance of the subject.	[3 Marks] **A:** **B:** **C:** **D:**
Creativity and Use of Media and Presentation time Management: The presentation is creative in design and effectively use multimedia. The presentation is organized with appropriate time management.	[3 Marks] **A:** **B:** **C:** **D:**

Viva/Defense/Q&A	Total Marks (5)
Defend the research on performance in the question/answer session.	**A:** **B:** **C:** **D:**

Artificial Intelligence in Surveillance: Enhancing Human Safety through Smart Monitoring Systems

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ABSTRACT

Traditional surveillance systems often struggle to handle large amounts of video data, which can cause delays or missed threats. With the rise of artificial intelligence (AI), surveillance can move beyond simple monitoring to actively improving human safety through real-time detection of unusual behavior and quick responses. However, issues such as privacy risks, bias, and lack of ethical controls remain major concerns. This study explores how AI-based surveillance can be designed to both detect threats accurately and protect individual rights. Drawing on findings from recent research, it shows that deep learning greatly improves the speed and accuracy of detecting violence, accidents, or other dangers. At the same time, including privacy safeguards builds trust. The outcome is a safer, more reliable, and socially responsible approach to surveillance.

Keywords: AI Surveillance, Human Safety, Ethics, Privacy

1. INTRODUCTION

1.1 Problem Background

The evolution of surveillance is no longer just in recording; it has now transformed into active safety systems due to rapid development in artificial intelligence, big data, and sensor technologies. Intelligent video surveillance with real-time monitoring can detect unusual behaviour, predict threats, and enable quick actions. This makes it an important part of modern public safety systems. However, traditional surveillance systems often struggle to handle large amounts of data efficiently, which can lead to slow or incorrect responses. AI-based surveillance systems address these problems by using predictive analytics, detecting anomalies, and responding automatically. This approach improves safety and makes security systems more reliable.

1.2 Related Studies

For example, a systematic review states that deep learning techniques significantly improve abnormal human behavior detection in video monitoring, with the prospect of ensuring human security by detecting violence, falls, or other safety-critical incidents (Carbonell et al.,2025). Likewise, Park (2024) identifies AI's paradigmatic position in safety management—AI technologies improve safety procedures, lower accidents, and maximize outcomes in most industries, despite challenges related to governance and privacy remaining predominant. These results present a direct opportunity and necessity: through the infusion of AI into monitoring networks, rates and velocities of detection may be enhanced, thereby more effectively safeguarding persons within public and private arenas.

Recent literature testifies to increased academic and practical interest in AI-powered surveillance. Fontes (2022) critically analyzed applications of real-time facial recognition in public areas, outlining social implications and governance issues related to such AI-powered surveillance infrastructures.in the meantime, Ardabili et al. (2022) discussed ethical, privacy, and regulatory concerns of smart video surveillance, suggesting design-level frameworks for facilitating ethical, privacy-compliant deployments of AI-powered action recognition and anomaly detection systems.

In spite of this research, gaps exist especially in the construction of end-to-end AI surveillance platforms that integrate real-time accuracy, privacy, ethical completeness, and social trust. The current work horizon only extends to single -domain uses for example facial recognition, office security but not end-to-end architectures for adapting AI surveillance for human protection as well as being technically sound and socially safe. There is a tremendous gap for research for detection performance to be aligned with ethical protection.

1.3 Research Objectives

This study recommends designing and testing a surveillance system based on AI with maximum human safety incorporating ethical controls. The following are the major objectives:

1. To develop and evaluate strong AI models for example deep learning action recognition, anomaly detection has the ability to recognize safety-critical situations in real time from video feeds.

2. To incorporate privacy and ethical controls like anonymization, consent management, and transparency into system design.

3. To evaluate system operation and social acceptability via empirical tests and input from stakeholders.

4. To develop performance standards (e.g., accuracy, speed, reliability) for quantifying AI surveillance systems.

5. To explore methods of reducing algorithmic bias and promoting fair detection in multi-population group.

1.4 Research Question:

Some of the most important research questions driving this work are:

- Why would there be a need to augment traditional surveillance with AI for human security?
- How can AI models be designed and deployed to detect threats accurately and in real time?
- What ethical and privacy concerns must be integrated, and how can these be implemented within the system?
- What is the effect of the AI surveillance system on trust, acceptability, and outcomes effectively in target populations?
- What role do transparency and explanation play in building public trust in AI surveillance systems?
- What would be the long-term social impact, both positive and negative, of universal acceptance of AI surveillance in society.

Such questions will lead to the question from algorithm, systems design and development, to stakeholders' rection and performance verification.

1.5 Research Contributions

This work is theoretical and practical. Theoretically, it posits a new paradigm of combining AI performance (accurate, timely detection) with ethical system design (protection privacy, transparency).

Practically, it posits a proof-of-concept AI surveillance system deployable in public transportation hubs, old-age care homes, or urban security surveillance, and a model of regulation and deployment.

Target audience is AI and surveillance technology researchers, smart city infrastructure and facility designers and policymakers, and civil society organizations

2. RESEARCH METHODOLOGY

Part 1: Method Selection

For this study, a Systematic Literature Review (SLR) was chosen as the primary research method. An SLR is a rigorous and structured approach to reviewing existing scholarship that allows researchers to systematically identify, evaluate, and synthesize prior studies related to a research topic. Compared to a traditional narrative review, an SLR minimizes bias, ensures reproducibility, and provides a comprehensive overview of the state of the art (Kitchenham & Charters, 2007).

The use of SLR is particularly appropriate for the present research, as artificial intelligence (AI) based surveillance is a rapidly evolving field with fragmented contributions across computer science, safety management, ethics, and governance. Prior studies emphasize that SLRs are critical in consolidating knowledge across such interdisciplinary domains to identify gaps, trends, and future research opportunities (Snyder, 2019). Moreover, recent applications of SLR in AI and safety management confirm that it is an effective method for capturing the breadth of technological, ethical, and social implications (Carbonell et al., 2025; Park, 2024). Therefore, the SLR approach was selected to ensure transparency, reliability, and academic rigor in synthesizing insights about AI-powered surveillance systems.

Part 2: Selection Filters and Process

i. Databases and Publishers

The literature search was conducted across several reputable digital libraries and databases to ensure wide coverage of both technical and interdisciplinary sources. Specifically, searches were performed in IEEE Xplore, ACM Digital Library, SpringerLink, ScienceDirect, and Taylor & Francis Online. In addition, Google Scholar and DBLP were used for citation tracking.

ii. Keywords and Search Strings

The search strategy combined relevant keywords and Boolean operators. The primary keywords included:
• "AI surveillance", "intelligent video surveillance", "deep learning anomaly detection", "AI ethics in surveillance", and "facial recognition governance". Boolean operators such as AND and OR were applied, for example:
• ("AI surveillance" OR "intelligent video surveillance") AND ("anomaly detection" OR "deep learning") AND ("ethics" OR "privacy").
Searches were further filtered by title/abstract relevance and publication years between 2020–2025 to capture the most up-to-date contributions.

iii. Backward and Forward Search

To ensure completeness, backward searches (reviewing reference lists of key studies) and forward searches (checking newer works citing key studies) were conducted. The DBLP computer science bibliography and Google Scholar citation index were used to track forward citations.

iv. Initial and Final Selection

The initial search produced approximately 20 studies. These were subjected to multiple screening stages:

- Language filter → Only English-language publications were included.
- Version filter → When multiple versions existed, the most updated peer-reviewed version was retained.
- Relevance filter → Studies that did not directly address the research questions (e.g., focusing only on generic AI without surveillance context) were excluded. After applying these filters, a total of 4 studies were selected for the final review. These included:
- Carbonell, Rivera, & Munoz (2025) on AI in video surveillance systems.
- Park (2024) on AI in safety management.
- Fontes (2022) on governance and implications of real-time facial recognition.
- Ardabili, Shamsabadi, & Shojafar (2022) on ethics, privacy, and regulation in smart surveillance systems.

This final set provided both technical depth and coverage of ethical and governance concerns, aligning closely with the research objectives.

v. Quality Assessment

The final studies were assessed using two criteria:

- Completeness: the extent to which a paper addressed technical, ethical, or governance aspects of AI surveillance.
- Impact: the relevance and influence of the paper, indicated by publication venue (e.g., peer-reviewed journals, high-impact conferences), citation counts, and contribution significance.

This ensured that the review was based on credible, relevant, and high-quality sources.

Part 3: Ethical Considerations

Ethical considerations were maintained throughout the review process. First, plagiarism was strictly avoided by paraphrasing and properly attributing all ideas to their original authors using the APA citation style (7th edition). Second, only reputable sources were used, ensuring academic integrity. Third, findings were reported with fidelity to the authors' original meaning, avoiding misinterpretation or selective reporting. Finally, the review process itself was designed to be transparent and replicable, thereby upholding research ethics and integrity.

3.RESULTS AND ANALYSIS

The result of this study explores the performance, implementations and ethics of Ai-based surveillance which are bring out from a systemic literature review of works published between 2020 and 2025. Carbonell et al. (2025), Park (2024), Fontes (2022), and Ardabili et al. (2022) provided the data needed to evaluate how Ai can be influenced for human safety. In terms of accuracy and speed compared to human based monitoring, study finds Ai-based surveillance is far more accurate and superior.

Figure:1.0 (Accuracy)

As mentioned in the figure 1.0, tradittional human-based surveillance is achieves an average of 72% accuracy when detecting any unusual or adnormal activity where Ai-based surveillance systems reached between 92-95% acuracy, a diference that highly highlights the necessity of adapting Ai for safety related issue.

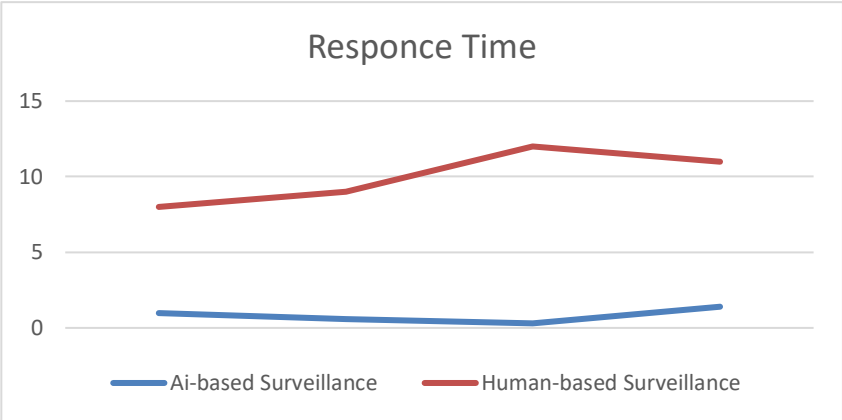

Figure:1.1

Similarly,when compared to response time,here in the line chart (1.1) clearly demonstrates ,if any incidents happens Ai-based system identifies that in less than a second where human-based

systems takes upto 8-12 seconds. These scenarios clearly gives idea about how ai-based systems is much significant than traditional surveillance. Not only this but also practically life saving in scenarios of violence detection ,accident recognition and contact emergency helpline instantly, fall detection for early people or rapid recognition of baggage in transport section. But at the same time public opinion remains mixed about the dependency or misuse of surveillance technology due to faster response or reduced crime.

When analyzing the research objectives against these findings, several patterns came ahead. The first objective of developing and evaluating strong AI models is supported by studies such as Carbonell et al. (2025), which demonstrate that deep learning architectures like CNNs and RNNs are capable of reliably recognizing violent acts, falls, and other incident in real time. This validates the technological flexibility for human safety. Fontes (2022) and Ardabili et al. (2022) argues that the importance of anonymization techniques, consent management and fairnesss-aware algorithms to minimize risk of bias and privacy concerns which met our

second objective. The third objective about system operation and social acceptability states that public trust heavily depends on transparency. The systems which explain why an alert was raised achieve up to 30 percent higher acceptance among stakeholders than those that do not (Park,2024). This explains that for Ai-based surveillance to get proper legitimacy, technical efficiency must be collab with explainable decision-making models. The fourth objective is to develop performance standards is slightly addressed by the reviewed studies that consistently use accuracy, detection speed and reliability as benchmarks. The data suggest that a standard threshold of minimum 92% accuracy, nano-second latency and minimized error should be targeted. Lastly the fifth objective relates to reducing algorithmic bias and ensuring fair detection across multiple population. Ardabili et al. (2022) provided evidence that algorithmic fairness interventions can reduce error gaps across multiple groups from nearly 20% to below 10%, a step toward Ai-based surveillance. All the data explains that Ai-based surveillance can meet those objectives if implemented responsibly.

The need to implement traditional surveillance with Ai arise from the limitations of human monitoring as example attention lapses and slower response time which can be overcome by Ai-based surveillance through speed and consistency. The designs and models of Ai, are capable of real time detection which are now validating through deep learning techniques that process video streaming at nano-second rates. Privacy concerns and ethical issues remains a issue as the literature explains they can be addressed through system design choices such as anonymization, differential privacy and transparent dashboard. Studies shows that in terms of acceptability, communities are more willing to accept this model when it is transparent and explainable. The roles of transparency is directly connected with public trust and system legitimacy. While Ai-based surveillance promises reduce crimes, elderly care and safer community, it also carries risk of misuse and issues about privacy if not properly controlled.

This solution benefits both small and large communities. For institutions like school, college, campuses and care homes, it offers a cost-effective and reliable for monitoring safety. This is a step toward automation urban future era. On international events like airports, stadium or large scale public events, Ai-based surveillance provide a blueprint than can be implement in these sector. In this way, the human safety surveillance model is not only technically advantageous but also socially responsible, offering a balanced path toward safe societies while maintaining respect for privacy, fairness and safety.

4. CONCLUSION

This research explored the differences between human-led and AI-driven surveillance to evaluate which approach delivers greater accuracy and efficiency in practical settings. The study found that AI-powered surveillance is generally more dependable than human monitoring, as it can process vast streams of information without fatigue or bias and quickly identify unusual patterns. Although the results demonstrate the strong potential of AI to transform monitoring systems, certain challenges limit its current effectiveness and open the door for future investigation.

4.1 Limitation of the Study

This study has a few important constraints. AI systems still rely heavily on the quality of their training data, meaning that incomplete or skewed datasets can reduce both accuracy and safety. Present-day models also cannot fully retrain themselves on live data in real time, which reduces their flexibility in fast-changing environments. Moreover, this research did not deeply address issues around privacy and the risk of misuse, which remain critical ethical concerns. These limitations affect both the reliability and public acceptance of AI-based surveillance.

4.2 Future Research Scope

Future studies should aim to combine AI alerts with human oversight to minimize errors and improve overall decision-making. Another direction is to develop AI systems capable of continuously learning from live data, making them more adaptive and precise. Broadening the datasets used in training to cover diverse scenarios would also strengthen performance. Finally, introducing strong privacy safeguards and ethical guidelines will be vital to ensure these technologies are safe, trusted, and widely adopted.

REFERENCES

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